

verters require a means to vary their output voltage in response to changes in their load.

As the PWM controller switches between two states ('on' and 'off') and has a soft-start circuit, the output power is proportional to 'on' and 'off' durations of the controller. The PWM controller IC provides a very high efficiency and allows regulatory compliance with relevant standards for harmonic emission. It also provides pulse-by-pulse current-limiting, under-voltage lockout, 7mA operating current (typical), soft-start on/off control, overload protection, over-current protection and over-voltage protection.

Digital power conversions

Elaborating power requirements of complex systems, Anant says, "As the complexity of the system climbs up the ladder, with as many as five voltage rails per processor, and with multiple processors per printed circuit board, power supply complexity goes up." These systems need tracking, sequencing and margining, coupled with tighter voltage regulations on voltage rails as low as 0.9V. Digital power was then thought as an effective alternative to address these needs.

A digital power controller provides system control functions that require multiple outputs, coordinated load sharing, hot-swap capability or extensive fault handling. This makes it suitable for future applications like wireless base stations, high-speed data, voice and video networks, and renewable energy. Digital features can be combined to create power systems that can accommodate requirements of today's most sophisticated systems as well as those of the future. Primarion, Zilker Labs, ADI, Fairchild Semiconductor, Microchip and Texas Instruments (TI) are some of the players in digital controllers.

Supervisory and sequencing ICs

Many applications today embed multiple functionalities. These functions

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work on different levels of voltages and need different types of voltage regulations. Anant explains, "Going back to our example of a cellphone, there are at least 13-14 different voltage rails inside a cellphone. So you have a single Lithium ion battery of 3.6V, supplying negative 18V biasing for its LCD, 5V for its processor and may be 12V to some other part of the circuit." You need to provide power to all the sub-systems in specified limits and specified sequence.

Supervisory ICs ensure that the system power supplies stay within the specified voltage threshold, by continuously comparing them with the power supply voltage. If the power source reaches that threshold, the supervisory IC generates a pulse that resets the system processor.

As multiple functional circuits are getting power-up and power-down, ICs usually face the problem of latching up. Power-supply sequencing solves the problem of avoiding latch-up effects in ICs during power-up and power-down. "Multivoltage microprocessors usually require to have their I/O voltages before applying core voltage or vice versa. On the other hand, the processor should be up and running before powering up the graphics," explains Chalakkal.

"Earlier, supervising, sequencing and controlling were discrete solutions. With recent developments in semiconductors, all of these functions are combined into a power management IC," Chalakkal says. Every designer has his own sequencing requirement. The voltage and timing

are programmable depending on the requirements.

Industry support to designers

"In every design, there is enough and more scope to reduce power. It starts with calculating power budget, selecting components and integrating everything together to make a much more reliable system. Going further, if a designer selects the silicon based on cost which is less efficient, the challenge is how to harvest more power from the same lossy power supply," says Chalakkal.

To ease the component selection process in power management and power supply design, silicon manufacturers provide a Web tool for selection. By merely entering values like input voltage, output voltage, current and temperature, the software proves to be a fast and easy power management design tool. ADI, National Semiconductor, Texas Instruments and many other companies provide such a tool.

One step forward are the actual specifications. These help you in understanding what should be the transient response, noise figure, power supply rejection ratio and converter design efficiency. You can get the answers by referring to the application notes and reference design solutions.

As every designer has different voltage, tolerance, thermal, supervisory and sequencing requirements, silicon vendors provide customised evaluation boards, PCB mini cards, etc. This helps in creating a much more reliable power management design.

Many companies are working on programmable power technology. This power-saving technology optimises logic, DSP and memory blocks to maximise performance where needed while delivering the lowest power elsewhere in the design. ●

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